

Early Stopping as Spectral Regularization for Least Squares Regression

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Sonthalia, Lok, Rebrova, "*On regularization via early stopping for least squares regression*", submitted (2025).

Full-batch gradient descent / Landweber iteration

We observe

$$y = X\beta^* + \varepsilon, \quad X \in \mathbb{R}^{n \times p}.$$

We iterate

$$\beta_k = \beta_{k-1} - \frac{\eta_k}{n} X^T (X\beta_{k-1} - y).$$

Choosing the final iterate is not only a computational decision:

the stopping time is a regularization parameter.

The resulting loss is determined by

$$X^T X, \quad \{\eta_k\}, \quad \text{and the stopping time } \hat{k}.$$

Leading question

If we stop at iteration $\hat{k}$, what regularizer did we choose, and in what sense was $\hat{k}$ a good time to stop?

Part	Question
Intro / metric	What does “good stopping” mean for prediction risk?
Exact dynamics	What is the discrete-time GD path?
Ridge equivalence	What ridge/Tikhonov penalty did stopping choose?
Stopping time	How do signal, noise, ridge, spectrum, and schedule determine $\hat{k}$?
Experiments	Does the estimate track the observed optimum?

Training data

We observe examples

$$(z_i, y_i), \quad i = 1, \dots, n.$$

- z_i : raw input, e.g. image, signal, covariates.
- y_i : observed response.
- Ψ : feature map.

$$x_i = \Psi(z_i) \in \mathbb{R}^p, \quad X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix}.$$

Prediction language

A parameter vector β predicts by

$$\hat{y}_{\text{new}} = x_{\text{new}}^T \beta.$$

The reference vector β^* is the coefficient vector of the best linear predictor using the chosen features $x = \Psi(z)$.

On the training data, define the residual by

$$y = X\beta^* + \varepsilon.$$

The model is linear in the features, not necessarily linear in the raw input.

Prediction risk: parameter error seen through test data

Let

$$h = \beta - \beta^*.$$

For a fresh test feature vector x_{new} , the prediction error is

$$x_{\text{new}}^T \beta - x_{\text{new}}^T \beta^* = x_{\text{new}}^T h.$$

Excess test error / prediction risk

$$R(\beta) = \mathbb{E}_{\text{test}} \left[(x_{\text{new}}^T (\beta - \beta^*))^2 \mid X \right] = (\beta - \beta^*)^T \Sigma_{\text{test}} (\beta - \beta^*).$$

What is Σ_{test} ?

$$\Sigma_{\text{test}} = \mathbb{E}_{\text{test}} [x_{\text{new}} x_{\text{new}}^T].$$

It weights which parameter errors matter for future prediction. If $\Sigma_{\text{test}} = I$, prediction risk is ordinary parameter error.

Conditioning on X is the fixed-design viewpoint: freeze the training geometry and average over fresh test inputs. We write $\Sigma = \Sigma_{\text{test}}$. This is where possible train–test covariance shift enters: the dynamics use $X^T X$, while the metric uses Σ .

Same iterate, different error metrics

Metric	Formula	Typical role
Training / residual error	$\frac{1}{n} \ y - X\beta\ _2^2$	Data fit; discrepancy principle; can be small when fitting noise.
Recovery / parameter error	$\ \beta - \beta^*\ _2^2$	Recover the unknown signal.
Prediction / excess test risk	$\ \beta - \beta^*\ _\Sigma^2$	Predict well on a fresh test point.

Note:

The same iterate can be excellent for prediction and poor for recovery, or vice versa. This talk optimizes the prediction-risk metric.

Why we care about prediction risk

Training error can be misleading

In modern regimes,

$$p \gtrsim n \text{ or } p \gg n,$$

the model may interpolate:

$$\|y - X\beta\| \approx 0.$$

But fitting the training labels does not answer whether the model predicts well on new inputs.

ML phenomena

- **Benign overfitting:** fit noise, still generalize.
- **Grokking / late generalization:** the model fits the data very quickly, but keeps learning for a while before it generalizes well.
- **Double descent:** test risk can be non-monotone.

So the stopping question changes

Not only:

When is the residual small?

But:

When is prediction risk minimized?

We study

$$R(\beta_k) = \mathbb{E}_{\text{test}} \left[(x_{\text{new}}^T \beta_k - x_{\text{new}}^T \beta^*)^2 \mid \mathcal{X} \right].$$

Here β^* is the best linear predictor in the chosen features.

Some related literature

1. Iterative regularization / inverse problems

- Semi-convergence, discrepancy principle; stopping often calibrated to recovery or residual.
- Engl–Hanke–Neubauer (1996);
Blanchard–Hoffmann–Reiß (2018); Hucker–Reiß (2025); Hansen (2025).

2. Hybrid Krylov and Bayesian regularization

- Generalized Tikhonov, hyperparameter choice via evidence / marginal posterior / scalable Krylov tools.
- Chung–Saibaba (2017);
Chung–Saibaba–Brown–Westman (2018);
Chung–Miller–Sabaté Landman–Saibaba (2025).

3. ML generalization phenomena

- Benign overfitting, double descent, grokking / late generalization.
- Bartlett–Long–Lugosi–Tsigler (2020);
Hastie–Montanari–Rosset–Tibshirani (2022);
Levi–Beck–Bar-Sinai (2024).

4. Early stopping, ridge, and gradient dynamics

- Early stopping as implicit regularization; gradient flow vs discrete GD; connections to ridge, risk dynamics, and optimal stopping estimates.
- Yao–Rosasco–Caponnetto (2007);
Raskutti–Wainwright–Yu (2014);
Ali–Kolter–Tibshirani (2019);
Advani–Saxe–Sompolinsky (2020); Shen–Gao–Ma (2022); Stark–Steinerberger (2025).

In this paper/talk: while specialized inverse-problem solvers are often more efficient; here full-batch GD is kept simple to expose the implicit regularizer and prediction-risk stopping rule.

What we want to understand

Known:

Early stopping acts like regularization:

stop early $\approx$ use an L_2 penalty.

Questions:

- ❶ **What is the stopped iterate?** Can we write β_k exactly for discrete GD and arbitrary step sizes?
- ❷ **What regularizer did we choose?** Is early stopping really scalar ridge, or a more general Tikhonov penalty?
- ❸ **When should we stop?** Can we estimate the stopping time that minimizes prediction risk?

What we show, in general:

Early stopping is an exact *spectral filter*; at time T , it corresponds to a generalized ridge penalty, and prediction risk gives a target learning budget.

Explicit optimization comparison

For ridge/Tikhonov parameter $\mu \geq 0$, consider

Ridge/Tikhonov least squares

$$L_\mu(\beta) = \frac{1}{2n} \|y - X\beta\|_2^2 + \frac{\mu}{2} \|\beta\|_2^2.$$

Gradient descent with arbitrary step sizes is

$$\beta_k = \beta_{k-1} - \eta_k \left[\frac{1}{n} X^T (X\beta_{k-1} - y) + \mu\beta_{k-1} \right].$$

We study the whole path β_k , not just its limit.

The empirical spectrum is the coordinate system

Let

$$X^T X = V \Lambda V^T, \quad X = U S V^T.$$

Work in the eigenbasis:

$$\tilde{\beta}_k = V^T \beta_k, \quad \tilde{\beta}^* = V^T \beta^*, \quad \tilde{\epsilon} = U^T \epsilon.$$

Key note:

Gradient descent decouples across the eigendirections of $X^T X$.

Technical idea:

one vector iteration $\longrightarrow$ many scalar recursions

How does the dynamic appear

Fix one eigenvalue j , and write

$$\zeta_j = \mu + \frac{\Lambda_j}{n}, \quad a_{k,j} = 1 - \eta_k \zeta_j.$$

From the GD update,

$$\tilde{\beta}_{k,j} - \tilde{\beta}_j^* = a_{k,j}(\tilde{\beta}_{k-1,j} - \tilde{\beta}_j^*) + \frac{\eta_k}{n} b_j,$$

where

$$b_j = (S^T \tilde{\epsilon})_j - \mu n \tilde{\beta}_j^*.$$

After two steps,

$$\tilde{\beta}_{2,j} - \tilde{\beta}_j^* = a_{2,j} a_{1,j} (\tilde{\beta}_{0,j} - \tilde{\beta}_j^*) + \frac{1}{n} (\eta_2 + a_{2,j} \eta_1) b_j.$$

The product that keeps appearing

$$\Phi_j(k; \mu) = \prod_{i=1}^k \left(1 - \eta_i \left(\mu + \frac{\Lambda_j}{n} \right) \right).$$

$$\Phi_j(k; \mu) = \prod_{i=1}^k \left(1 - \eta_i \left(\mu + \frac{\Lambda_j}{n} \right) \right), \quad \Phi(k; \mu) = \text{diag}(\Phi_j(k; \mu)).$$

- $\Phi_j(k; \mu)$ is the fraction of the initial error still left in eigendirection j after k steps.
- Large Λ_j : the mode is learned quickly.
- Small Λ_j : the mode is learned slowly.

The entire step-size schedule enters through this product, which makes arbitrary schedules tractable.

The two-part split

Thm 1: exact discrete trajectory

For fixed X , arbitrary initialization, arbitrary step sizes, and any $\mu \geq 0$, the full-batch GD iterate has an exact spectral representation:

$$\beta_k = V\Phi(k; \mu)V^T\beta_0 + (I - V\Phi(k; \mu)V^T)(\mu nI + X^TX)^\dagger X^Ty.$$

From the trajectory formula,

$$\tilde{\beta}_k - \tilde{\beta}^* = \underbrace{\Phi(k; \mu)(\tilde{\beta}_0 - \tilde{\beta}^*)}_{\text{unlearned signal}} + \underbrace{(\mu nI + \Lambda)^\dagger(I - \Phi(k; \mu))(S^T\xi - \mu n\tilde{\beta}^*)}_{\text{fitted noise and ridge bias}}.$$

Early iterations: Large unlearned-signal term; little noise has been fit.

Late iterations: Signal is learned, but the fitted-noise term can dominate prediction risk.

Like in semi-convergence works, training moves from initialization toward the ridge solution, but at different speeds in different empirical eigendirections.

Risk formula: the same split in the test norm

Assume

$$\mathbb{E}[\varepsilon \mid X] = 0, \quad \mathbb{E}[\varepsilon \varepsilon^T \mid X] = \tau^2 I_n.$$

For $\mu = 0$,

$$\mathbb{E}_\varepsilon \left[\|\beta_k - \beta^\star\|_\Sigma^2 \mid X \right] = \underbrace{\left\| \Sigma^{1/2} V \Phi(k) (\tilde{\beta}_0 - \tilde{\beta}^\star) \right\|_2^2}_{\text{unlearned signal}} + \underbrace{\tau^2 \left\| \Sigma^{1/2} V (I - \Phi(k)) S^\dagger \right\|_F^2}_{\text{fitted noise}}.$$

The iteration is governed by $X^T X$, but the risk is measured in the test norm Σ .

Technical note

If $A_k := \Sigma^{1/2} V (I - \Phi(k)) S^\dagger U^T$, then

$$\mathbb{E}_\varepsilon \|A_k \varepsilon\|_2^2 = \text{Tr}(A_k^T A_k \mathbb{E}[\varepsilon \varepsilon^T \mid X]) = \tau^2 \|A_k\|_F^2.$$

The cross term vanishes because $\mathbb{E}[\varepsilon \mid X] = 0$. No Gaussian or sub-Gaussian tail assumption is used here.

Comparison: semi-convergence in inverse problems

Classical inverse-problems picture

For noisy data $b = A\bar{u} + e$, iterative regularization uses spectral filters in the singular-vector basis.

$$u_k - \bar{u} = \underbrace{\text{unrecovered signal}}_{\downarrow} + \underbrace{\text{amplified noise}}_{\uparrow}.$$

Semi-convergence: improve first, then deteriorate as noise enters unstable directions.

What changes here?

The algebra is analogous:

$$\tilde{\beta}_k - \tilde{\beta}^* = \text{unlearned signal} + \text{fitted noise}.$$

But the stopping criterion is prediction risk,

$$\|\beta_k - \beta^*\|_{\Sigma}^2,$$

not only recovery error or residual discrepancy.

Discrepancy-principle comparison: Classical discrepancy rules stop when the residual reaches the noise scale. Here residual size alone need not track test risk, especially when interpolation is possible.

Learning-rate schedules: accumulated budget

For eigendirection j , set

$$\zeta_j = \mu + \Lambda_j/n, \quad \Phi_j(k) = \prod_{i=1}^k (1 - \eta_i \zeta_j), \quad B_k := \sum_{i=1}^k \eta_i.$$

The exact object is the product filter $\Phi_j(k)$. For stable small steps,

$$\log \Phi_j(k) = \sum_{i=1}^k \log(1 - \eta_i \zeta_j) = -\zeta_j B_k + O\left(\zeta_j^2 \sum_{i=1}^k \eta_i^2\right).$$

B_k is a first-order time budget for the exact filter. The statistics will give a target budget B^* , and the schedule converts that budget into an iteration number.

Note:

If $B_\infty = \sum_i \eta_i < \infty$, the schedule may never reach B^* . Then it simply does not train far enough to cross the predicted signal-noise threshold.

Learning-rate schedules: budgets and exact formulas

Budget behavior

$$\eta_i = \eta : B_k = \eta k,$$

$$\eta_i = \eta/i : B_k \sim \eta \log k,$$

$$\eta_i = \eta/i^m : B_k \rightarrow B_\infty \quad (m > 1).$$

Constant steps give linear progress; harmonic steps keep learning slowly; fast decay has finite total budget.

Exact filters

The budget B_k is a first-order summary, but the product filter is often computable.

For constant steps,

$$\Phi_j(k) = (1 - \eta \zeta_j)^k.$$

For exponential decay, $\eta_i = \eta_0 q^{i-1}$,

$$\Phi_j(k) = (\eta_0 \zeta_j; q)_k.$$

Here $(a; q)_k = \prod_{\ell=0}^{k-1} (1 - a q^\ell)$ is the q -Pochhammer symbol.

Gamma-function formulas also appear for polynomial/additive decay; cyclic schedules give products over periods.

Early stopping is closer to generalized ridge than scalar ridge

Generalized ridge/Tikhonov solves

$$\min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|y - X\beta\|_2^2 + \frac{1}{2} \|D\beta\|_2^2.$$

Scalar ridge

$$D = \sqrt{\mu}I$$

One penalty for all directions.

Generalized ridge

D diagonal in the V -basis

Different penalties for empirical eigendirections.

Main idea

Early stopping usually does not behave like scalar ridge; it behaves like spectral ridge.

What shrinkage does early stopping apply?

Take the ridgeless case $\mu = 0$, zero initialization $\beta_0 = 0$, and

$$X^T X = V \Lambda V^T.$$

Let

$$\beta_{\text{mn}} := (X^T X)^\dagger X^T y = V \Lambda^\dagger V^T X^T y$$

be the minimum-norm least-squares solution.

From the exact trajectory formula,

$$\beta_T = V(I - \Phi(T)) \Lambda^\dagger V^T X^T y.$$

Then, in each trainable eigendirection,

$$\tilde{\beta}_{T,j} = (1 - \Phi_j(T)) \tilde{\beta}_{\text{mn},j}.$$

Interpretation

Stopping time T decides how much of each empirical mode is allowed through.

Choose the generalized ridge penalty to match shrinkage

Let the generalized ridge penalty be diagonal in the V -basis:

$$D = \text{diag}(d_j) V^T.$$

The generalized ridge solution is

$$\beta^{(D)} = (X^T X + n D^T D)^\dagger X^T y.$$

In direction j ,

$$\tilde{\beta}_j^{(D)} = \frac{\Lambda_j}{\Lambda_j + n d_j^2} \tilde{\beta}_{\text{mn},j}.$$

Match generalized ridge to early stopping:

$$\frac{\Lambda_j}{\Lambda_j + n d_j^2} = 1 - \Phi_j(T) \quad \implies \quad d_j^2 = \frac{\Lambda_j}{n} \frac{\Phi_j(T)}{1 - \Phi_j(T)}.$$

The exact dictionary: stopping time $\leftrightarrow$ spectral penalty

Thm 2: stopped GD equals generalized ridge

Assume $\mu = 0$ and $\beta_0 = 0$. For any stopping time T , the stopped iterate β_T is the minimum-norm generalized ridge/Tikhonov solution with

$$D = \left(\frac{1}{n} \Lambda \Phi(T) (I - \Phi(T))^\dagger \right)^{1/2} V^T.$$

In one spectral direction

$$d_j^2 = \frac{\Lambda_j}{n} \frac{\Phi_j(T)}{1 - \Phi_j(T)}.$$

The stopping time T chooses a direction-dependent penalty.

For stable monotone steps, $0 \leq \Phi_j(T) < 1$, so $d_j^2 \geq 0$.

From the risk formula to a stopping rule

Let

$$P = V^T \Sigma V, \quad g = \tilde{\beta}_0 - \tilde{\beta}^*, \quad \Phi = \Phi(k).$$

For $\mu = 0$, the previous risk formula can be written as

$$R_\varepsilon(\beta_k \mid X, g) = \underbrace{g^T \Phi P \Phi g}_{\text{unlearned signal}} + \underbrace{\tau^2 \text{Tr}(\Lambda^\dagger (I - \Phi) P (I - \Phi))}_{\text{fitted noise}}.$$

Noise part

Since $\Lambda^\dagger$ and Φ are diagonal,

$$\tau^2 \text{Tr}(\Lambda^\dagger (I - \Phi) P (I - \Phi)) = \tau^2 \sum_{\Lambda_j > 0} P_{jj} \frac{(1 - \Phi_j)^2}{\Lambda_j}.$$

This is already modewise.

Signal part

But

$$g^T \Phi P \Phi g = \sum_{i,j} P_{ij} \Phi_i \Phi_j g_i g_j.$$

So the signal term depends on the unknown alignment of $\beta_0 - \beta^*$ with the empirical eigenspaces.

Note: to get a clean stopping rule, we need to remove this unknown alignment dependence.

Generic-signal average: what the “prior” does

To average over unknown signal orientation, assume

$$\mathbf{g} = \tilde{\beta}_0 - \tilde{\beta}^\star, \quad \mathbb{E}_{\mathbf{g}}[\mathbf{g}] = 0, \quad \mathbb{E}_{\mathbf{g}}[\mathbf{g}\mathbf{g}^T] = \sigma^2 \mathbf{I}.$$

Then the non-scalar signal term becomes scalar by trace:

$$\begin{aligned} \mathbb{E}_{\mathbf{g}} \mathbf{g}^T \Phi P \Phi \mathbf{g} &= \text{Tr}(\Phi P \Phi \mathbb{E}_{\mathbf{g}}[\mathbf{g}\mathbf{g}^T]) \\ &= \sigma^2 \sum_j P_{jj} \Phi_j(k)^2. \end{aligned}$$

Write $R_B(\beta_k) := \mathbb{E}_{\mathbf{g}} R_{\varepsilon}(\beta_k \mid X, \mathbf{g})$. In the ridgeless trainable modes,

$$R_B(\beta_k) = \sum_{\Lambda_j > 0} P_{jj} \left[\sigma^2 \Phi_j(k)^2 + \frac{\tau^2}{\Lambda_j} (1 - \Phi_j(k))^2 \right] + \text{constant nullspace term}.$$

What assumptions are used where?

Result	Assumptions	Meaning
Exact trajectory	fixed X , arbitrary spectrum, arbitrary $\{\eta_k\}$	deterministic spectral algebra
Generalized ridge equivalence	same; plus $\mu = 0$, $\beta_0 = 0$	stopping time T chooses a spectral penalty
Expected risk formula	$\mathbb{E}[\varepsilon \mid X] = 0$, $\mathbb{E}[\varepsilon\varepsilon^T \mid X] = \tau^2 I$	τ^2 is residual-noise strength
Stopping threshold	isotropic signal average with variance σ^2 , plus monotone differentiable filters	σ^2 is initial-error signal strength; average over alignment
Global $\hat{k}$	replace Λ_j by $\bar{\Lambda}$	representative-eigenvalue heuristic

Modewise risk balance

Thm 3: stopping threshold

Under the assumptions on the previous slide, each trainable mode $\Lambda_j > 0$ changes from decreasing to increasing risk contribution when the filter crosses

$$\Phi_j^* = \frac{\tau^2}{\tau^2 + \sigma^2 \Lambda_j} \quad (\mu = 0).$$

With explicit ridge and zero initialization, the threshold becomes

$$\Phi_j^*(\mu) = \frac{\tau^2 - \sigma^2 \mu n}{\tau^2 + \sigma^2 \Lambda_j}.$$

Immediate consequence

Let

$$\mu^* = \frac{\tau^2}{n\sigma^2}.$$

If $\mu \geq \mu^*$, then $\Phi_j^*(\mu) \leq 0$: explicit ridge already supplies enough regularization in this risk criterion.

When is early stopping beneficial?

(Ridgeless case: $\mu = 0$) Assume the noise moment condition, the spherical signal average, and stable monotone filters:

$$0 \leq \Phi_j(k) \leq 1, \quad \Phi_j(k) \text{ nonincreasing in } k.$$

For each trainable mode, define the critical filter value

$$\Phi_j^* = \frac{\tau^2}{\tau^2 + \sigma^2 \Lambda_j}.$$

Theorem 8: beneficial

If, for every trainable mode,

$$\lim_{k \rightarrow \infty} \Phi_j(k) < \Phi_j^*,$$

then the risk eventually increases. So there is a finite time T after which training is worse:

$$R_B(\beta_k) \geq R_B(\beta_T) \quad (k \geq T).$$

Theorem 9: not beneficial

If, for every trainable mode,

$$\lim_{k \rightarrow \infty} \Phi_j(k) \geq \Phi_j^*,$$

then the trajectory never passes the noise-fitting threshold. Further training keeps decreasing Bayes risk, so early stopping is not beneficial.

Ridge case

For the ridge case, the derivative can be written in the compact form

$$\partial_k R_B(\beta_k) = 2 \sum_j P_{jj} \partial_k \Phi_j(k) \frac{\Lambda_j(\tau^2 + \sigma^2 \Lambda_j)}{(\mu n + \Lambda_j)^2} [\Phi_j(k) - \Phi_j^*(\mu)].$$

Since stable training gives

$$\partial_k \Phi_j(k) \leq 0,$$

the sign is easy to read.

Before the threshold: beneficial

If

$$\Phi_j(k) > \Phi_j^*(\mu),$$

the unlearned-signal term is still too large, and training decreases risk in that mode.

After the threshold: not beneficial

If

$$\Phi_j(k) < \Phi_j^*(\mu),$$

the fitted-noise / ridge-bias tradeoff dominates, and further training increases risk in that mode.

Stopping is the moment when learning signal and fitting noise balance.

From threshold to target budget

Assume $\mu < \mu^*$, so $\Phi_j^*(\mu) > 0$. The per-mode target is

$$\Phi_j(k) \approx \Phi_j^*(\mu).$$

Using

$$\Phi_j(k) \approx \exp\left[-\left(\mu + \frac{\Lambda_j}{n}\right) B_k\right], \quad B_k = \sum_{i=1}^k \eta_i,$$

we get

$$B_j^* \approx \frac{\log\left(\frac{\tau^2 + \sigma^2 \Lambda_j}{\tau^2 - \sigma^2 \mu n}\right)}{\mu + \Lambda_j/n}.$$

Interpretation

Statistics gives a target budget B_j^* ; the learning-rate schedule converts budget into iteration number.

If $B_\infty < B_j^*$, this mode never reaches the predicted threshold, so there is no finite predicted stopping time for that mode.

From spectral targets to one global stopping time

A single GD trajectory cannot choose all Φ_j independently. The modewise formula gives many targets. For one scalar stopping estimate, replace Λ_j by a representative eigenvalue

$$\bar{\Lambda} = \frac{\text{Tr}(X^T X)}{\text{rank}(X^T X)}.$$

Define

$$B^* := \frac{\log\left(\frac{\tau^2 + \sigma^2 \bar{\Lambda}}{\tau^2 - \sigma^2 \mu n}\right)}{\mu + \bar{\Lambda}/n}.$$

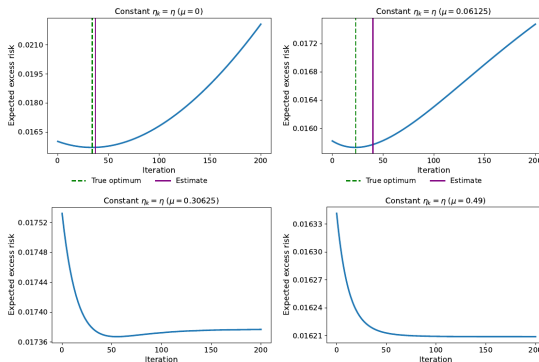
Then, if the set is nonempty,

$$\hat{k} = \min\{k : B_k \geq B^*\}.$$

The displayed $\hat{k}$ is a representative-eigenvalue heuristic; it can be crude for strongly multiscale spectra. If $B_k < B^*$ throughout the horizon, report no finite $\hat{k}$ and train to the horizon/convergence.

Explicit ridge vs implicit regularization

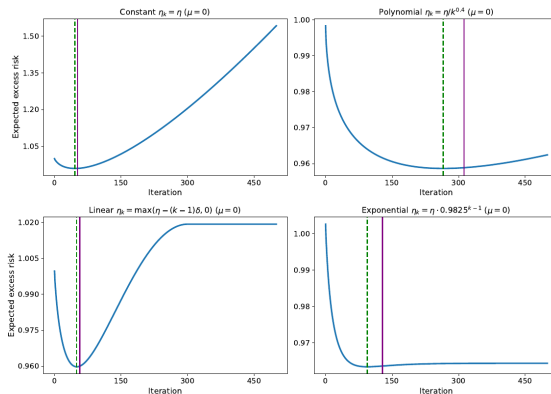
Synthetic experiment comparing $\mu < \mu^*$, $\mu = \mu^*$, and $\mu > \mu^*$.



For $\mu < \mu^*$, early stopping helps. Near or above μ^* , explicit ridge has already done the regularization. Green – true optimum, purple – our estimate. Here, $n = 40$, $p = 100$, diagonal covariance $\Sigma_{jj} = j^{-2}$, averaged over 800 spherical signals $\beta^* \sim \mathcal{N}(0, p^{-1}I)$ and Gaussian noise with $\tau = 0.35$, so $\sigma^2 = 0.01$ and $\mu^* = \frac{\tau^2}{n\sigma^2} = 0.30625$. Constant step size $\eta = 0.01/(\Lambda_1/n)$,

Schedules change the budget-to-iteration map

MNIST feature matrix; same stopping estimate applied across schedules.



Different schedules can have similar risk behavior at different iteration numbers because they accumulate budget differently. Green – true optimum, purple – our estimate.

Conclusions

Discrete GD is exactly analyzable

For least squares, full-batch GD is controlled by the product filter

$$\Phi_j(k; \mu) = \prod_{i=1}^k \left(1 - \eta_i \left(\mu + \frac{\Lambda_j}{n} \right) \right).$$

This is a discrete-time statement, so arbitrary learning-rate schedules remain visible.

Stopping is spectral regularization

The stopped iterate is not usually scalar ridge. It is generalized ridge/Tikhonov with a direction-dependent penalty in the empirical eigenbasis.

Stopping is calibrated by prediction risk

The risk calculation turns this into a signal–noise threshold, then into a target budget B^* , and finally into $\hat{k}$ through the schedule.

Main message: early stopping is a spectral regularizer, and the schedule is the budget-to-iteration map.

Future directions and questions

- **SGD and adaptive methods.** Not covered here.

Related: Lok–Sonthalia–Rebrova, *Discrete error dynamics of mini-batch gradient descent for least squares regression*, ALT 2025 / arXiv:2406.03696. Mini-batch dynamics are already substantially more complex; early-stopping question remains open.

- **Beyond fixed features.** Nonlinear feature learning – ?
- **Spectral versus scalar stopping.** The global $\hat{k}$ heuristic is scalar, while the true optimum is eigendirection-dependent.
- **Data-driven calibration.** Can signal/noise strength be estimated more robustly, or replaced by a fully data-driven rule?

Thank you!

Thank you!

Sonthalia, Lok, Rebrova,
"On regularization via early stopping for least squares regression", submitted (2025).
<https://erebrova.github.io/EarlyStopping2.pdf>

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